

localhost:8081/lab/workspaces/auto-H/tree/comp40771_lab9

File

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/ comp40771_lab9 /

Name	Modified
lab9	yesterday
comp40771_lab9.i...	1h ago
README.md	yesterday
subscriber.py	yesterday

Launcher

comp40771_la

Notebo

Python 3
(ipykernel)

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Python 3
(ipykernel)

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Other

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localhost:8081/lab/workspaces/auto-H/tree/comp40771_lab9/lab9

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/ comp40771_lab9 / lab9 /

Name	Modified
src	21h ago
tests	yesterday
Jenkinsfile	yesterday
LICENSE	yesterday
README.md	yesterday

Launcher

comp40771_lab9/lab9

Notebook

Python 3
(ipykernel)

Console

Python 3
(ipykernel)

Other

localhost:8081/lab/workspaces/auto-H/tree/comp40771_lab9/lab9/tests/test_generated_code.py

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Launcher comp40771_lab9.ipynb test_generated_code.py

/ ... / lab9 / tests /

Name	Modified
placeholder	yesterday
test_generated_co...	yesterday

```
1 import importlib.util
2 import sys
3 from pathlib import Path
4
5 REPO_ROOT = Path(__file__).resolve().parents[1]
6 SRC_DIR = REPO_ROOT / "src"
7
8
9 def load_module(module_path: Path):
10     spec = importlib.util.spec_from_file_location(module_path.stem, module_path)
11     module = importlib.util.module_from_spec(spec)
12     sys.modules[module_path.stem] = module
13     spec.loader.exec_module(module)
14     return module
15
16
17 def get_python_files():
18     return sorted(SRC_DIR.glob("*.py"))
19
20
21 def test_original_file_exists():
22     assert (SRC_DIR / "original_monitor.py").exists()
23
24
25 def test_all_scripts_load():
26     for script in get_python_files():
27         module = load_module(script)
28         assert module is not None
29
```

localhost:8081/lab/workspaces/auto-H/tree/comp40771_lab9/lab9/src/original_monitor.py

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Launcher comp40771_lab9.ipynb test_generated_code.py

/ ... / lab9 / src /

Name	Modified
improved_deepco...	21h ago
improved_deepco...	21h ago
improved_phi3_mi...	23h ago
improved_phi3_mi...	23h ago
improved_qwen2_...	21h ago
improved_qwen2_...	21h ago
original_monitor.py	yesterday
placeholder	yesterday

```
1 import os
2 import time
3 import tracemalloc
4 from pathlib import Path
5
6
7 def list_all_paths(root="/opt/data"):
8     root_path = Path(root)
9     results = []
10
11     if not root_path.exists():
12         print(f"Path does not exist: {root}")
13         return results
14
15     for current_root, dirs, files in os.walk(root):
16         for d in dirs:
17             results.append(os.path.join(current_root, d))
18         for f in files:
19             results.append(os.path.join(current_root, f))
20
21     return sorted(results)
22
23
24 def collect_memory_usage(duration_seconds=1.0, interval_seconds=0.1):
25     tracemalloc.start()
26     samples = []
27
28     start = time.time()
29     while time.time() - start < duration_seconds:
30         current, peak = tracemalloc.get_traced_memory()
```

localhost:8081/lab/workspaces/auto-H/tree/comp40771_lab8/comp40771_lab8.ipynb

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comp40771_lab8 /

Name	Modified
comp40771_lab8.i...	8d ago
harness.py	8d ago
README.md	8d ago

Launcher

comp40771_lab8.ipynb

Markdown

Lab 8

Baseline Centralised Learning and Distributed Robustness Assessment (CL-DR)

Establish a standard Centralised Learning (CL) model and evaluating its behaviour in both benign and adversarial distributed conditions. Simulate a controlled binary classification task, after which a simple neural classifier is trained using conventional gradient-based optimisation not only under normal test conditions but also against Fast Gradient Sign Method (FGSM) perturbations to measure adversarial robustness. The harness is subsequently used to simulate Non-Independent and Identically Distributed (Non-IID) data across multiple nodes, enabling the use of indicators such as mean client accuracy and instability (variance across clients). Communication Cost (CC) is approximated via parameter c proxy. Overall, CL-DRA establishes the baseline performance, robustness, and system-level variability against which subsequent optimisations can be evaluated.

```
[1]: import math, random, time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import torch
import torch.nn as nn
import torch.nn.functional as F

import optuna # Part C

# import your script
import harness
```

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Container memory usage ⓘ

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☐	lab7	-	-	-	0%	12 minutes ago	☐ ⋮
☐	monitor	6149cd50388a	lab7-monitor		0%	5 days ago	☐ ⋮
☐	api	df6f55476bdf	lab7-api	8000:8000	0%	5 days ago	☐ ⋮
☐	ollama	ee73730169aa	lab7-ollama	11434:11434	0%	12 minutes ago	☐ ⋮
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☐	monitor	6149cd50388a	lab7-monitor		0%	5 days ago	
☐	api	df6f55476bdf	lab7-api	8000:8000	0%	5 days ago	
☐	ollama	ee73730169aa	lab7-ollama	11434:11434	0%	14 minutes ago	
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☐	jenkins-comp40771	96eade07f5e7	lab9-jenkins	50001:50000	0.34%	14 minutes ago	
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☐	eb949347415e_ollama	6c7ff6171938	lab9-ollama	11434:11434	0%	1 second ago	
☐	comp40771_container	1097bfc00d85	pedrombmachado/nest:cc	8081:8080	102.48%	8 seconds ago	